8. Father of a family wishes to divide his square field bounded by $x=0$, $x=4$, $y=4$ and $y=0$ along the curve $y^2=4x$ and $x^2=4y$ into three equal parts for his wife, daughter and son. Is it possible to divide? If so, find the area to be divided among them.
9. The curve $y=(x-2)^2+1$ has a minimum point at *P*. A point *Q* on the curve is such that the slope of *PQ* is 2. Find the area bounded by the curve and the chord *PQ*.
10. Find the area of the region common to the circle $x^2+y^2=16$ and the parabola $y^2=6x$.

9.9 Volume of a solid obtained by revolving area about an axis

Definite integrals have applications in finding volumes of solids of revolution about a fixed axis. By a solid of revolution about a fixed axis, we mean that a solid is generated when a plane region in a given plane undergoes one full revolution about a fixed axis in the plane. For instance, consider the semi circular plane region inside the circle $x^2+y^2=a^2$ and above the *x*-axis. See Fig.9.34.

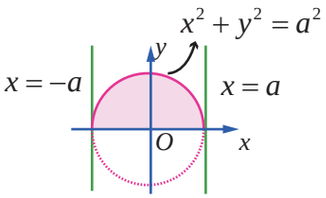


Fig. 9.34

If this region is given one complete rotation (revolution for $360°=2\pi$ radians) about *x*-axis, then a solid called a sphere is generated.

In the same manner, if you want to generate a right-circular cylinder with radius *a* and height *h*, you can consider the rectangular plane region bounded between the straight lines $y=0$, $y=a$, $x=0$ and $x=h$ in the *xy*-plane. See Fig.9.35. If this region is given one complete rotation (revolution for $360°=2\pi$ radians) about *x*-axis, then a solid called a cylinder is generated.

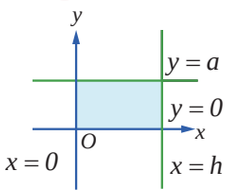


Fig. 9.35

We restrict ourselves to obtain volume of solid of revolution about *x*-axis or *y*-axis. Whenever solid of revolution about *x*-axis is considered, the plane region that is revolved about *x*-axis lies above the *x*-axis. So, in this region $y\geq 0$. Whenever solid of revolution about *y*-axis is considered, the plane region that is revolved about *y*-axis lies to the right of *y*-axis. So, in this region $x\geq 0$. We shall find the formula for finding the volume of the solid of revolution of the plane region in the first quadrant bounded by the curve $y=f(x)$, *x*-axis and the lines $x=a$ and $x=b>a$ about *x*-axis. **The derivation of the formula is based upon the formula that the volume of a cylinder of radius *r* and the height *h* is $\pi r^2 h$.**

Assume that every line parallel to *y*-axis lying between the lines $x=a$ and $x=b>a$ cuts the curve $y=f(x)$ in the first quadrant exactly at one point. Divide $[a,b]$ into *n* segments by $x_1, x_2, ..., x_{n-1}$ such that

$$a=x_0<x_1<x_2<...<x_{n-1}<x_n=b,\ x_i-x_{i-1}=\Delta x=\frac{b-a}{n},\ i=1,2,...,n.$$

For each $i=0,1,2,...,n-1$, the region in the *xy*-plane between the ordinates at x_i and $x_i+\Delta x$ which lies between the *x*-axis and the curve $y=f(x)$ can be **approximated** to an infinitesimal rectangle having area $y_i\Delta x$, where $y_i=f(x_i)$. When the plane region bounded by the curve $y=f(x)$, *x*-axis, and lines $x=a$ and $x=b$ is rotated by $360°$ about *x*-axis, each of the infinitesimal rectangles at $x=x_i$ also revolves and generates an elementary solid which is **approximately** a thin cylindrical disc with radius y_i and height Δx. See Fig.9.36. The volume of the cylindrical disc at

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